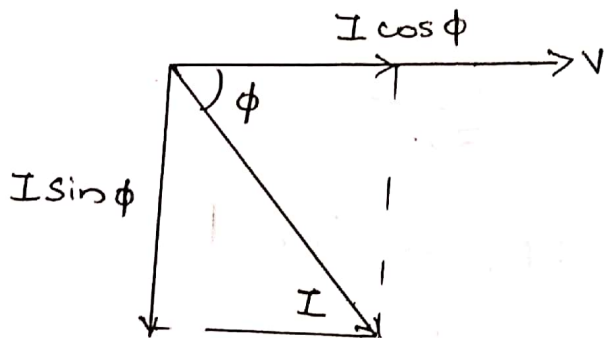


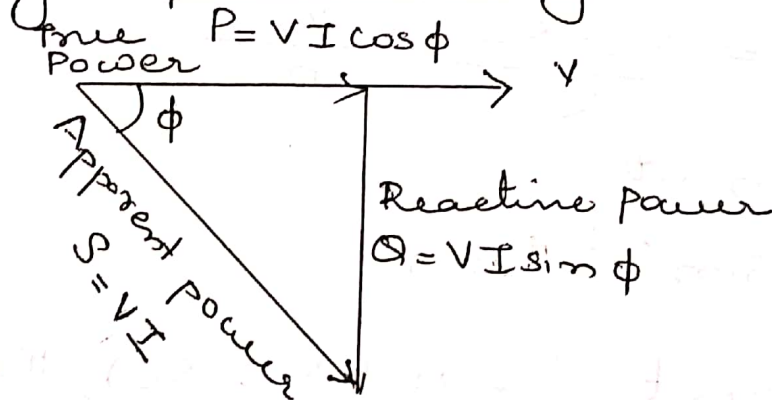
Apparent, True and reactive power

In an inductive circuit, current I lags behind the voltage V by ϕ . The current I can be resolved into two components.

- i) $I \cos \phi$ in phase with V
- 2) $I \sin \phi$, 90° out of phase with V



If we multiply each of the current phases by V we get power triangle



1) Apparent power - The total power that appears to be transferred between the source and load

- $S = VI$ (Product of applied voltage and current)
- Unit - volt amperes.

1. True power - Power which is actually consumed in the circuit is called true power or active power. Current in phase with voltage produces true power. It is the useful component of apparent power. $I \cos \phi$ is called useful component

$$P = VI \cos \phi \quad \text{Unit - Watts}$$

$$P = VI \cos \phi \quad [V = IZ, \cos \phi = R/Z]$$

$$= IZ \cdot I \frac{R}{Z} = I^2 R \quad [\text{Active power is power consumed in resistance}]$$

3. Reactive power - The component of apparent power which is neither consumed nor does any useful work.

$$Q = VI \sin \phi \quad \text{unit - volt amperes reactive (VAR)}$$

1. $I \sin \phi$ is called useless component.

$$\bullet S^2 = P^2 + Q^2$$

$$\bullet \cos \phi = \frac{\text{True power}}{\text{Apparent power}}$$

$$\bullet P = VI \cos \phi, \quad Q = VI \sin \phi$$

THREE PHASE SYSTEM

The a.c circuits which are termed single phase circuits contain a single alternating current and voltage wave.

A single phase generator producing a single phase supply has only one armature winding.

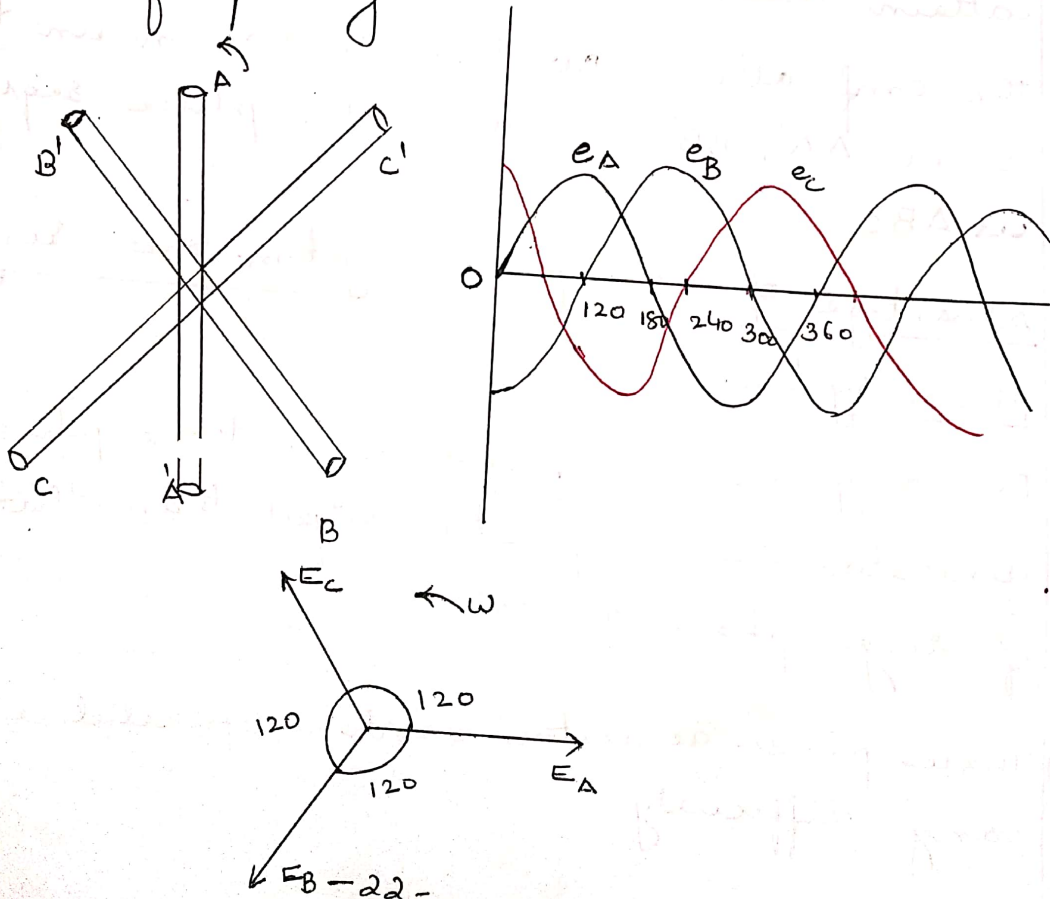
A two phase generator producing two phase supply has two windings displaced by 90° .

A three phase generator has 3 windings displaced by 120° .

Three phase system is most popular among polyphase systems

Generation of three phase voltages

Figure shows an elementary three phase alternator. The three identical coils A, B and C are symmetrically placed in such a way that emfs induced in them are displaced by 120° from one another. They should be kept under the influence of time varying magnetic field. Since the coils are identical and are subjected to the same rotating field, the emfs induced in them will be of same magnitude and frequency.



The equations of 3 emfs are

$$e_A = E_m \sin \omega t$$

$$e_B = E_m \sin (\omega t - 120^\circ)$$

$$e_C = E_m \sin (\omega t - 240^\circ)$$

Phase sequence

In a three phase system, there are 3 voltages having same magnitude and frequency displaced by an angle of 120° .

They are attaining their positive maximum in a particular order. The order in which the voltages (or emfs) in the three phases attain their maximum called phase sequence.

The emf attains positive maximum in the order AA' , BB' , CC' , thus the phase sequence is ABC .

Advantage of three phase system over single phase system

- 1) For a given size of frame, a three phase generator has a greater output than that of single phase generator.
- 2) Three phase generators work in parallel with any difficulty.

Three phase transmission line require lesser amount of conductor material for transmitting the same amount of power over a single phase line.

- 4) Three phase motors possess uniform torque whereas single phase motors possess a pulsating torque.
- 5) Poly phase induction motors are self starting whereas single phase induction motors are not self starting.

Symmetrical and balanced 3 ϕ system

A S/m is said to be symmetrical when the various voltages are equal in magnitude and are displaced from one another by equal angles.

The system is balanced when the various voltages are equal in magnitude, various currents are equal in magnitude and the phase difference between them are equal.

There are two methods of interconnecting 3 phases.

① Star connection

② Delta connection

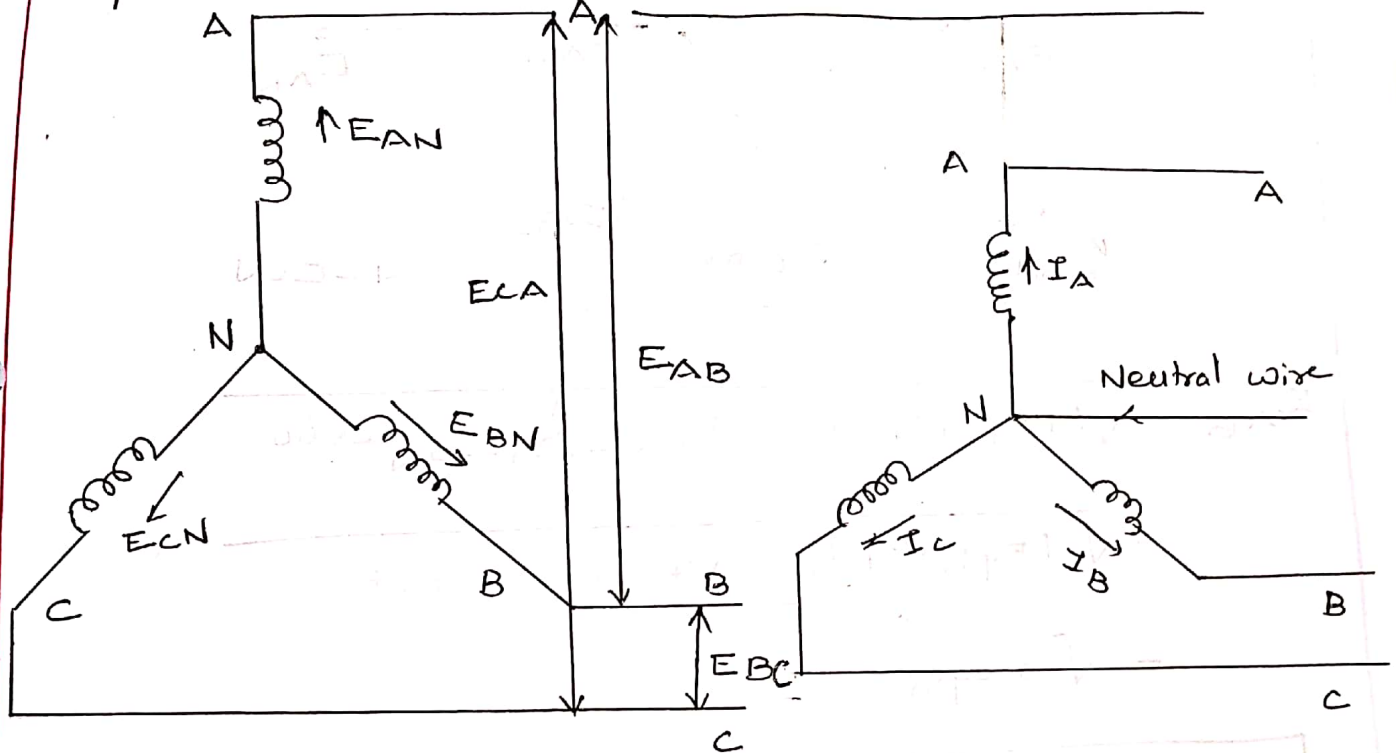
Three phases are joined together to form a common junction N called star or neutral point.

The three line conductors run from three free ends and are denoted as A, B and C. Voltage between any line and neutral point i.e. voltage across each winding is called phase voltage.

Voltage between any two lines is called line voltage.

currents flowing in the phase are called phase currents and those flowing in the lines are called line currents. Phase sequence is ABC.

STAR CONNECTION (Y)



The common point N may or may not be brought out. If a neutral conductor is present it is called a three phase four wire system. If neutral conductor absent - three phase three wire system.

Relation between line and phase voltages

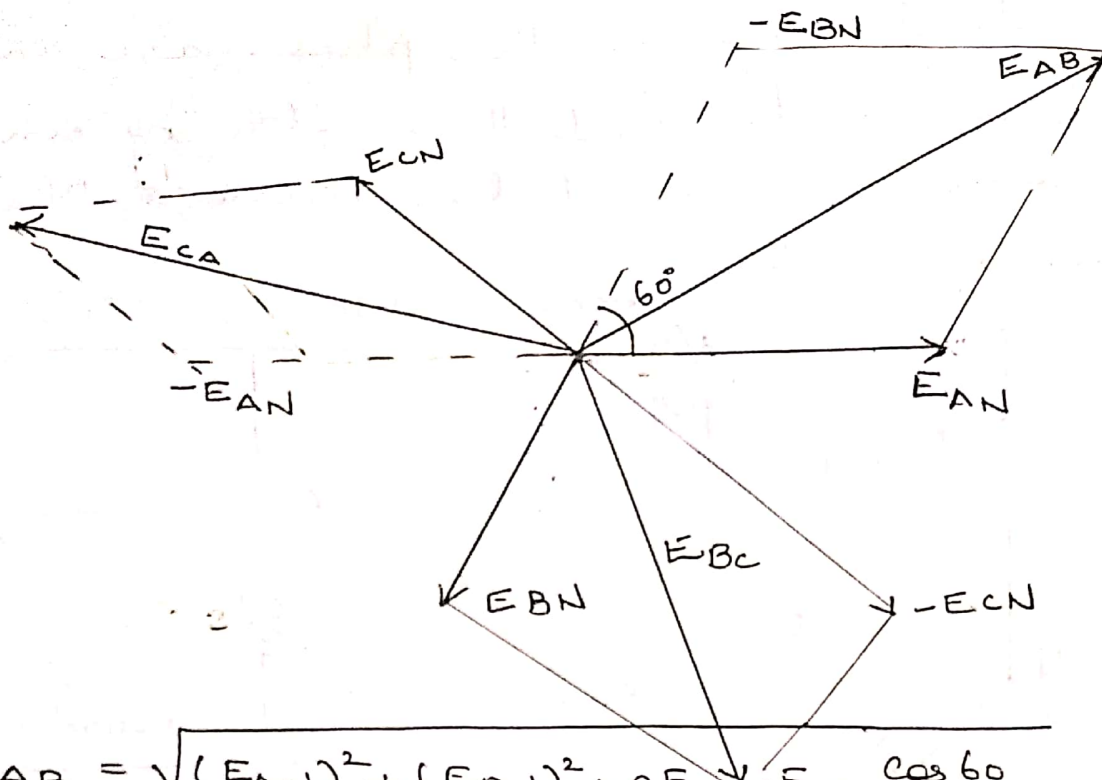
$$E_{AB} = E_{AN} - E_{BN}$$

$$E_{BC} = E_{BN} - E_{CN}$$

$$E_{CA} = E_{CN} - E_{AN}$$

E_{AN}, E_{BN}, E_{CN} - Phase Voltages

E_{AB}, E_{BC}, E_{CA} - line Voltages



$$\begin{aligned}
 E_{AB} &= \sqrt{(E_{AN})^2 + (E_{BN})^2 + 2E_{AN}E_{BN}\cos 60} \\
 &= \sqrt{(E_{ph})^2 + (E_{ph})^2 + 2E_{ph}^2\cos 60} \\
 &= \sqrt{3E_{ph}^2}
 \end{aligned}$$

$$E_L = \sqrt{3} E_{ph}$$

(line voltages are 30° ahead of their respective phase voltages)

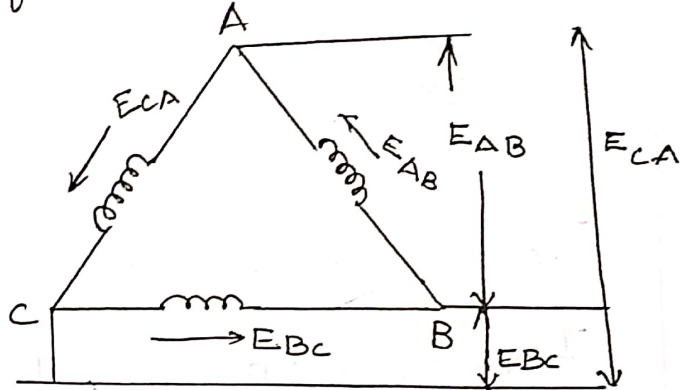
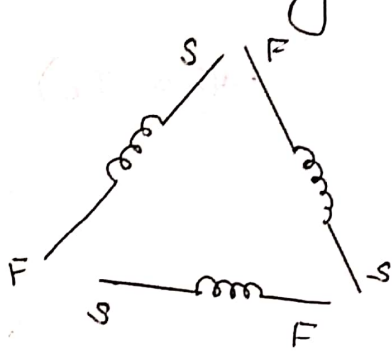
Relation between line and phase currents

In γ connection, each line conductor is in series to a phase.

$$\therefore I_L = I_{ph}$$

Delta or Mesh connection (Δ)

In this method of interconnection, the dissimilar ends of 3 phase windings are joined together i.e) finishing end of one phase is connected to the starting end of the other phase and so on.



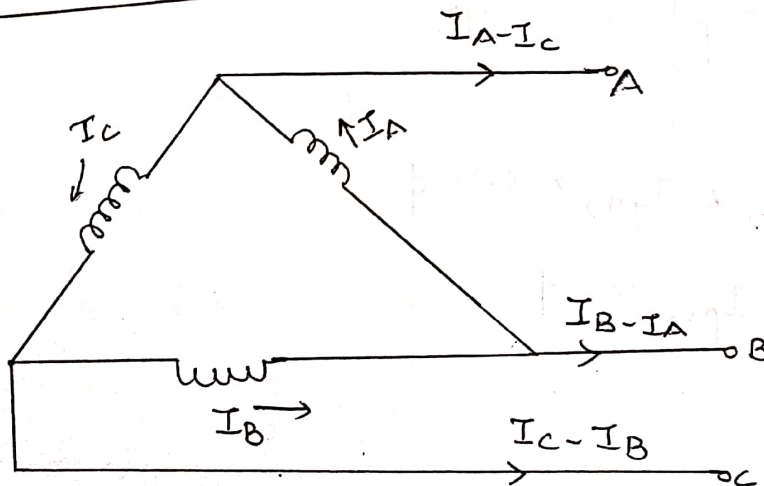
Relation between line and phase voltage

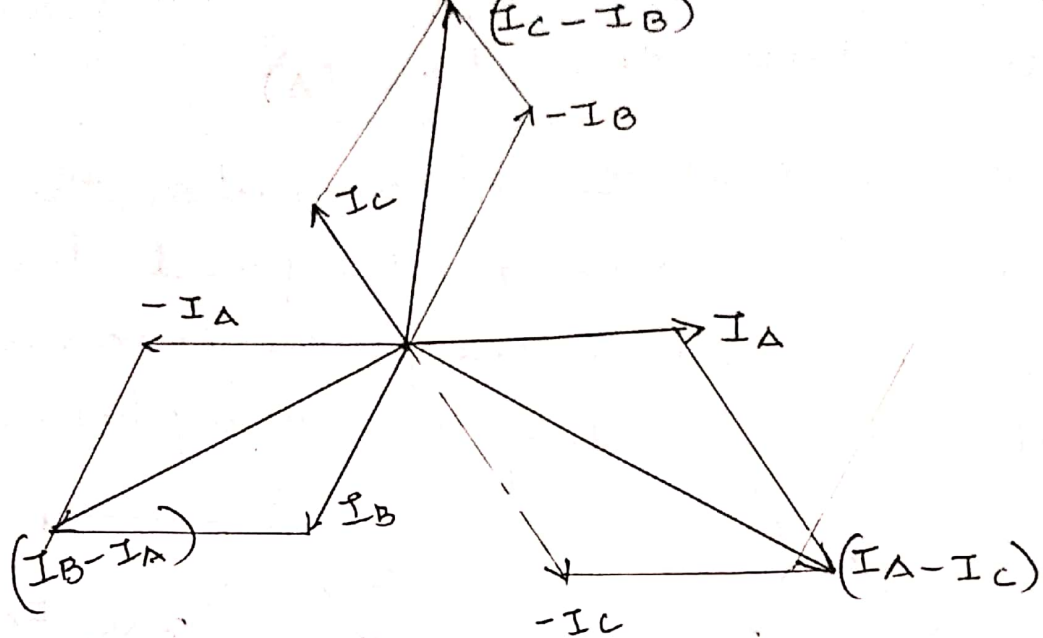
Since the system is balanced the three phase voltages are equal in magnitude but displaced by 120° from each other.

One phase winding is included between line

$$E_L = E_{ph}$$

Relation between line and phase currents





$$I_A = I_B = I_C = I_{ph}$$

$$I_A - I_C = I_B - I_C = I_C - I_B = I_L$$

$$I_L = \sqrt{(I_A)^2 + (I_B)^2 + 2 I_A I_B \cos 60}$$

$$= \sqrt{I_{ph}^2 + I_{ph}^2 + 2 I_{ph}^2 \cos 60}$$

$$= \sqrt{3} I_{ph}$$

$$I_L = \sqrt{3} I_{ph}$$

Power in 3 phase system

$P = 3 \times \text{power in each phase}$

$$= 3 \times E_{ph} \times I_{ph} \times \cos \phi$$

$$P = 3 E_{ph} I_{ph} \cos \phi$$

for γ connection

$$E_{ph} = \frac{E_L}{\sqrt{3}}, \quad I_{ph} = I_L$$

$$\underline{\underline{P = \sqrt{3} E_L I_L \cos \phi}}$$

for Δ connection

$$E_{ph} = E_L \quad I_{ph} = I_L / \sqrt{3}$$

$$P = 3 E_L \frac{I_L}{\sqrt{3}} \cos \phi$$

$$\underline{\underline{P = \sqrt{3} E_L I_L \cos \phi}}$$